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.NET & MICROSOFT TECHNOLOGIES

Two Developers, One Mission: Make a Test Database That Doesn't Suck

If you've ever been frustrated by a test database filled with nonsense, duplicates, sensitive information, or YOLO-style restored production backups, this session is for you.

Join Peter Kruis and Tonie Huizer, two energetic data enthusiasts, on an epic (and slightly chaotic) quest to build test databases that are safe, realistic, and actually useful for development and testing. With decades of combined experience, they'll take you on a journey through the wild world of test data, showing just how bad it can get, and how good it can be.

Through engaging role play, real-world examples, and live demo's, they'll expose the dangers of shady test environments and share smarter, safer ways to manage test data.

Audience participation is encouraged; because (bad) test databases are a team sport!

You'll leave entertained, but more importantly, you'll leave with tools and strategies to stop your test database from being the weakest link in your development chain.

Because let's face it: your test database shouldn't suck.



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SeaQL.nl
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Data, DevOps & SQL @ Sea!

17 & 18 April 2026, Ouddorp,
The Netherlands



New Mail

Urgent

Sender: RetroPixels

URGENT: Performance Issue in usp_SendPeriodDiscountEmails – Needs Fix TODAY

RetroPixels

To ● Peter Kruis

Cc ● BOSS



RetroPixels IT

Good morning,

We've identified a **critical performance issue** with the stored procedure `usp_SendPeriodDiscountEmails`. Execution currently takes around **20 seconds**, which is too long for queuing discount emails.

This is **urgent** — the discount codes need to be sent **TODAY**. Please prioritize optimizing this stored procedure.

Here are the parameters being used:

```
EXEC [dbo].[usp_SendPeriodDiscountEmails]
    @From = '2025-09-01',
    @To = '2025-10-01',
    @DiscountCode = N'123',
    @Status = 'PAID',
    @CountryCode = 'NL',
    @MinimumTotalGrand = 100,
    @ProductId = 21;
GO
```

Please ensure this is reviewed, optimized, and re-tested as soon as possible so the emailing process can proceed without further delay.

Thanks,
RetroPixels IT Team





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FW: URGENT: Performance Issue in usp_SendPeriodDiscountEmails – Needs Fix TODAY

RetroPixels

To ● Peter Kruis; ● BOSS



Reply



Reply All



Forward



wo 8/10/2025 13:56



Stupid discount action - [TODO: Replace subject in PRD]

Outlook item



Hello,

How can this happen?! We got complaints for our customers that they got an email like attached. We tracked the source of this email and seen it coming from your SMTP server. What happened! This will costs us customers!

@Boss, how are we going to solve this financially?

From: Peter Kruis <peter.kruis@monin-it.be>

Sent: woensdag 1 oktober 2025 12:22

To: Peter Kruis <peter.kruis@monin-it.be>

Cc: Peter Kruis <peter.kruis@monin-it.be>

Subject: URGENT: Performance Issue in usp_SendPeriodDiscountEmails – Needs Fix TODAY



RetroPixels IT

Good morning,

We've identified a **critical performance issue** with the stored procedure usp_SendPeriodDiscountEmails. Execution currently takes around **20 seconds**, which is too



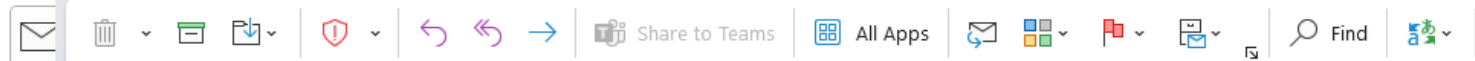
New Mail

Urgent

Sender: RetroPixels

FW: URGENT: Stupid discount action - [TODO: Replace subject in PRD] - Message (HTML) (Read-Only)

Retr
To File Message Help Tell me what you want to do



Hello,

How c
happe

@Bos

From:

Sent: v

To: Pel

Cc: Pel

Subje

Stupid discount action - [TODO: Replace subject in PRD]

RetroPixels
To Me

Reply

Reply

from your SMTP server. What



RetroPixels Discount!

Dear mr John King,

We've watched you while ordering, and therefor.... DISCOUNT FOR YOU! WOOOHOOOO. After that we will call you on +31 87 123 456

123
GO

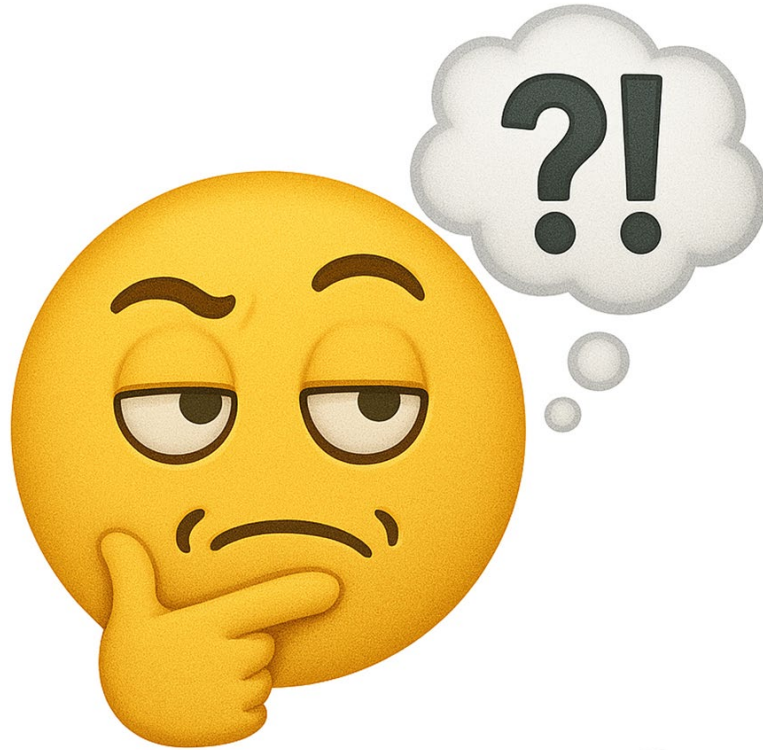
[CUSTOMER_FOOTER]

Thanks,
RetroPixels IT Team

Forward



wo 8/10/2025 13:56



What makes a good test database?



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- 
- **Developers**
 - **QA/Testers**
 - **Performance/Load engineer**
 - **Support engineer**
 - **DBAs**

What are the challenges?



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Common challenges

Challenge	Developer	QA/Tester	Performance/ Load engineer	Support Engineer	DBA
Data masking & privacy	✓	✓	✓	✓	✓
Size vs Performance	△	△	✓	△	✓
Applications Settings	✓	✓	△	✓	
Co-workers 🤔	✓	✓	✓	✓	✓
Reset, refresh & versioning	✓	✓	✓	✓	✓



Data masking & privacy

54% of organizations say they've already had a **breach or data theft involving sensitive data in non-production**; 86% admit they allow compliance exceptions in those environments; 91% are concerned about the expanding exposure footprint.

71% of enterprises copy real production data into development and testing environments, which are typically less protected and monitored; 41% have no clear approach for handling PII in those environments.

Sources:

<https://www.red-gate.com/blog/database-devops/the-risks-and-rewards-of-using-production-data-for-testing>

<https://www.datanami.com/this-just-in/delphix-2024-data-compliance-report-highlights-growing-risks-in-non-production-environments/>



Data masking & privacy

1. Classify (sensitive) data

Free Solutions	Paid Solutions
SQL Server Data Discovery & Classification	Redgate SQL Data Catalog Redgate Test Data Manager
DBATools.io	Delphix Test Data Management
	Microsoft Purview



DEMO Data Classification





Data masking & privacy

2. Mask data

Free Solutions

Custom scripting

DBATools.io

Paid Solutions

Redgate Data Masker
Redgate Test Data Manager

Delphix Test Data Management



DEMO

Data masking





Size vs Performance

- Big Database stretch **backup/restore** windows
- Full-size copies **inflate storage & infra costs**
- **Unnecessary** heavy datasets **slows down** builds/tests
- Sometime a **partial** database is not enough

Sources:

<https://learn.microsoft.com/en-us/sql/relational-databases/backup-restore/full-database-backups-sql-server?view=sql-server-ver17>
<https://martinfowler.com/articles/continuousIntegration.html>



Data virtualization

Free Solutions

Custom scripting

SQL Server Docker container
with read-only image layer

dbaclone

Paid Solutions

Redgate Clone

Windocks

(Pure)Storage snapshot



Subsetting

Free Solutions

Custom scripting

Jailer

Paid Solutions

Redgate Test Data Manager

DATPROF Subset

Delphix Test Data Management



DEMO

Subsetting





Application settings

- **Inconsistency**: settings diverge per **environment**
- Secret **leakage**: SMTP/API keys/payment/etc end up in **backups & clones**
- Unsafe **endpoints**: tests hit **real services** (just like we demo'd)



Application settings

Free Solutions	Paid Solutions
Custom scripting	Redgate Data Masker Redgate Test Data Manager
DBATools.io	Delphix Test Data Management
Relocating settings	



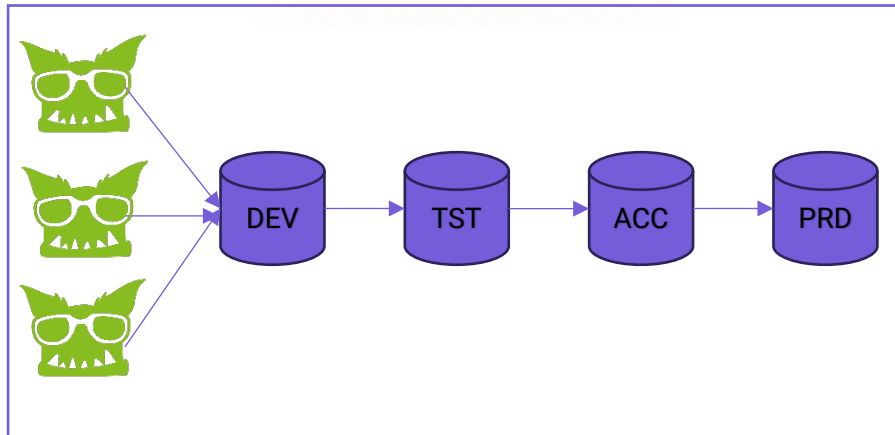
Relocation settings

- Separated (settings) database
- Application config files
- Migration / update script, after provisioning

⚠️ Co-workers?

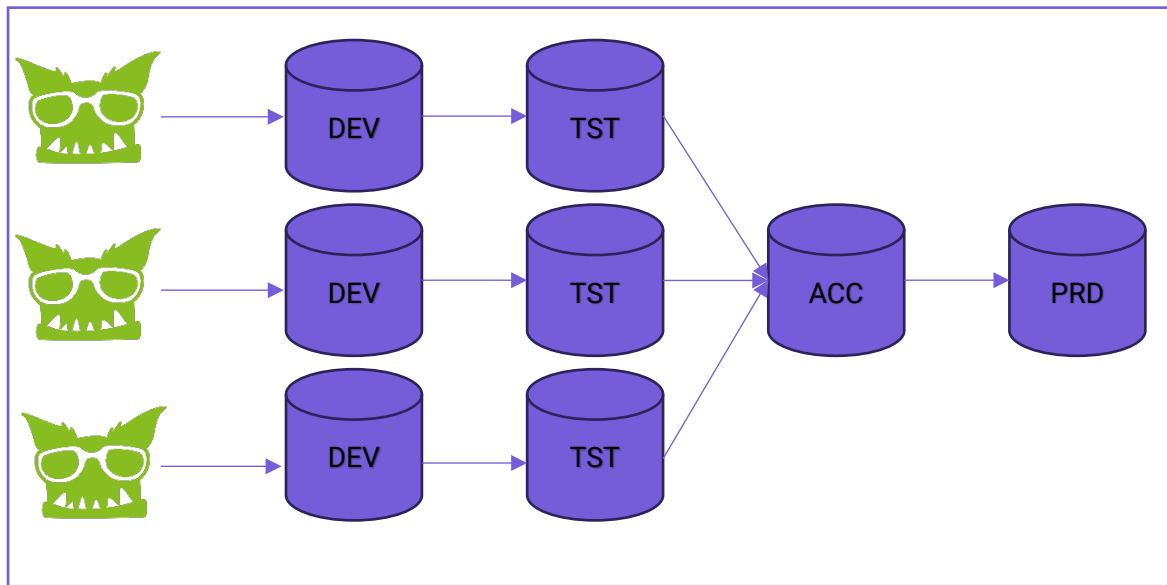


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Dedicated environment





Reset & Refresh

- **Test fails**, data corrupt: **new** database **needed**
- The **test** is **done**, who **cleans** up my **mess**?
- **No** or little **knowledge** of the process, but **fresh databases** are **needed**

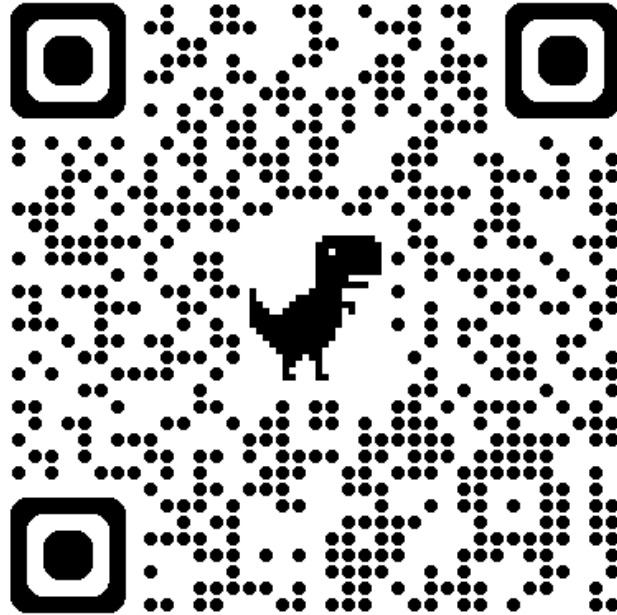


Automation

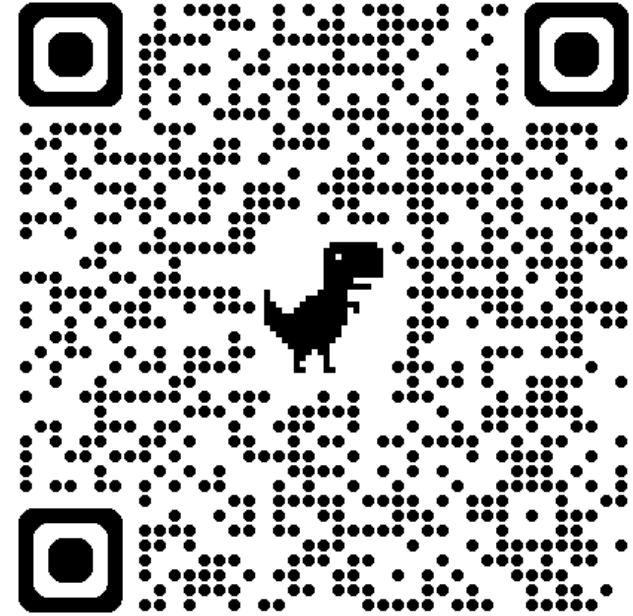
- Use **PowerShell**, but only that lives in **Git**
 - In a **separated** repo
- Use **pipelines**, but first...
 - Start **local**
 - Use verbose **logging**
 - Create **multiple** scripts



DEMO Automation



[Automation – Repos](#)

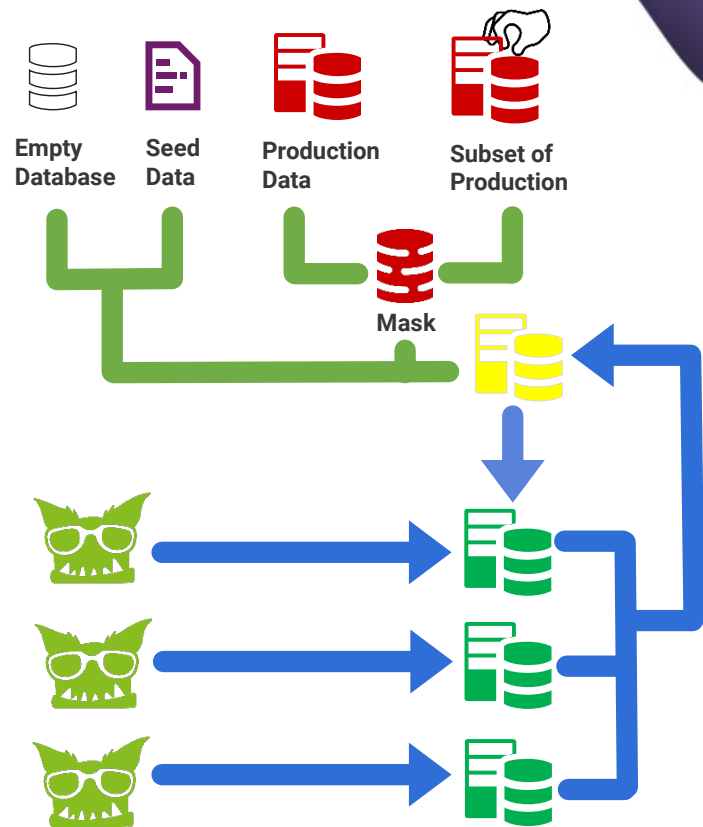


[Pipelines - Run](#)



So, how to create the test DB that doesn't suck...

- Choose a data set
 - Classify and mask this (when needed)
- Create golden images
- Spin up copies of this image
 - for every dev/test task
- Automate it
 - The creation
 - The disposal
 - The image refresh





Questions !?

Session Feedback 💖



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Doesn't Suck

Thanks for your attention!



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